Navid Bahadoran

Problem Set II

1. Classify the following models as $ARMA(p,q)$ (i.e., find the values of p and q. Be careful about parameter redundancy), and determine whether they are stationary and invertible.

   (a) $y_t = y_{t-1} + \varepsilon_t - \frac{1}{2}\varepsilon_{t-1} - \frac{1}{2}\varepsilon_{t-2}$,

   (b) $y_t = 2y_{t-1} - 2y_{t-2} + \varepsilon_t - \frac{3}{4}\varepsilon_{t-1}$,

   (c) $y_t = \frac{5}{4}y_{t-1} + \frac{3}{2}y_{t-2} + \varepsilon_t - 2\varepsilon_{t-1}$,

   where $\varepsilon_t \sim WN(0,1)$ in all processes.

(a)

$(1-L)y_t = \left(1 - \frac{1}{2}L - \frac{1}{2}L^2\right)\epsilon_t \Rightarrow (1-L)y_t = (1-L)\left(1 + \frac{1}{2}L\right)\epsilon_t \Rightarrow y_t = \left(1 + \frac{1}{2}L\right)\epsilon_t$

$\Rightarrow y_t = 1 + \frac{1}{2}\epsilon_t$It is ARMA(0,1) process which is always stationary and since the roots of $1 + \frac{1}{2}z = 0\ is, -2\ which\ is\ outside\ of\ unit\ circle, so\ it\ is\ invertible.$

(b)

$(1 - 2L + 2L^2)y_t = \left(1 - \frac{3}{4}L\right)\epsilon_t$ There is no common root, so it is ARMA(2,1).

$1 - 2z + 2z^2 = 0 \Rightarrow z = 0.5 \pm 0.5i$ which are inside the unit circle; therefore, it is not stationery. However, since the root of MA part is $\frac{4}{3}$, and it is outside of the unit circle, then it is invertible.

(c)

$\left(1 - \frac{5}{4}L - \frac{3}{2}L^2\right)y_t = (1-2L)\epsilon_t \Rightarrow \left(\frac{1}{2} - L\right)\left(2 + \frac{3}{2}L\right)y_t = (1-2L)\epsilon_t$

$\Rightarrow \left(2 + \frac{3}{2}L\right)y_t = 2\epsilon_t \Rightarrow \left(1 + \frac{3}{4}L\right)y_t = \epsilon_t$ Therefore the process is ARMA(1,0). Since the other root is $\frac{4}{3}$ which is out of unit circle, then it is stationary. Also it is invertible as we can write $\epsilon_t = y_t + \frac{3}{4}y_{t-1}$

2. Consider the following $ARMA\,(2,1)$ model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim WN\left(0, \sigma^2\right).$$

Show that the Yule-Walker equations are given by

$$\begin{aligned}
\gamma_0 &= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \left(\theta_1 \phi_1 + \theta_1^2 + 1\right) \sigma^2, \\
\gamma_1 &= \phi_1 \gamma_0 + \phi_2 \gamma_1 + \theta_1 \sigma^2, \\
\gamma_h &= \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}, \quad \text{for } h \geq 2.
\end{aligned}$$

$\gamma_0 = \mathbb{E}[y_t^2] = \mathbb{E}[y_t(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1})]$

$= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \mathbb{E}[\epsilon_t y_t] + \theta_1 \mathbb{E}[\epsilon_{t-1} y_t]$

$\mathbb{E}[\epsilon_t y_t] = \mathbb{E}[\epsilon_t(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1})] = \sigma^2$ (1)

$\mathbb{E}[\epsilon_{t-1} y_t] = \mathbb{E}[\epsilon_{t-1}(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1})] = \theta_1 \sigma^2 + \phi_1 \mathbb{E}[\epsilon_{t-1} y_{t-1}]$
$and\ \mathbb{E}[\epsilon_{t-1} y_{t-1}] = \mathbb{E}[\epsilon_{t-1}(\phi_1 y_{t-2} + \phi_2 y_{t-3} + \epsilon_{t-1} + \theta_1 \epsilon_{t-2})] = \mathbb{E}[\epsilon_{t-1}^2] = \sigma^2$ (2)
$\Rightarrow \mathbb{E}[\epsilon_{t-1} y_t] = (\theta_1 + \phi_1)\sigma^2$ (3)

$\Rightarrow \boxed{\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2 + \theta_1\left((\theta_1 + \phi_1)\sigma^2\right) = \phi_1 \gamma_1 + \phi_2 \gamma_2 + (1 + \theta_1^2 + \theta_1 \phi_1)\sigma^2}$

$\gamma_1 = \mathbb{E}[y_t y_{t-1}] = \mathbb{E}[y_{t-1}(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1})] = \phi_1 \gamma_0 + \phi_2 \gamma_1 + \theta_1 \mathbb{E}[\epsilon_{t-1} y_{t-1}]$

From using (2) we have:

$\boxed{\gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1 + \theta_1 \sigma^2}$

$\gamma_1 = \mathbb{E}[y_t y_{t-h}] = \mathbb{E}[y_{t-h}(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1})]$
$= \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2} + \underbrace{\mathbb{E}[\epsilon_t y_{t-h}]}_{0} + \underbrace{\theta_1 \mathbb{E}[\epsilon_{t-1} y_{t-h}]}_{0} = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2} \ \ for\ h \geq 2$

3. Consider the following $AR(2)$ process:

$$y_t = 3 + 0.3y_{t-1} + 0.04y_{t-2} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ is a Gaussian white noise process with variance 0.25. Suppose that $y_{99} = 4$ and $y_{100} = 3.5$.

(a) Compute the 1-step- and 2-step-ahead point forecasts and their standard errors at the forecast origin $t = 100$. Construct 95% prediction intervals for the forecasts.

(b) Compute the 100-step-ahead point forecast and its standard error at the forecast origin $t = 100$.

(a)

$\hat{y}_{101} = \mathbb{E}[y_{101}|\Omega_t] = 3 + 0.3y_{100} + 0.04y_{99} = 3 + (0.3 \times 3.5) + (0.04 \times 4)$
$\Rightarrow \hat{y}_{101} = 3 + 1.05 + 0.16 = 4.21$

$e_{101} = y_{101} - \hat{y}_{101} = 3 + 0.3y_{100} + 0.04y_{99} + \epsilon_{101} - 3 + 0.3y_{100} + 0.04y_{99}$
$\Rightarrow e_{101} = \epsilon_{101} \sim N(0, 0.25)$

0.95% prediction interval: $\hat{y}_{101} \pm 1.96 \times \sqrt{0.25} = 4.21 \pm 0.05 = 5.19 \; or \; 3.23$

$\hat{y}_{102} = \mathbb{E}[y_{102}|\Omega_t] = 3 + 0.3y_{101} + 0.04y_{100} = 3 + (0.3 \times 4.21) + (0.04 \times 3.5)$
$\Rightarrow \hat{y}_{102} = 3 + 0.657 + 0.14 = 4.403$

$e_{102} = y_{102} - \hat{y}_{102} = 3 + 0.3y_{101} + 0.04y_{100} + \epsilon_{102} - 3 + 0.3\hat{y}_{101} + 0.04y_{100}$
$\Rightarrow e_{102} = \epsilon_{102} + 0.3e_{101} = \epsilon_{102} + 0.3\epsilon_{101} \sim N\big(0, 0.25(1 + 0.09)\big) = N(0, 0.2725)$

0.95% prediction interval: $\hat{y}_{102} \pm 1.96 \times \sqrt{0.2725} = 4.403 \pm 1.023 = 5.43 \; or \; 3.38$

(b)

For large h (100 here), as we know the AR process is converging to unconditional mean which is:

$$\mu = \frac{\phi_0}{1 - \phi_1 - \phi_2} = \frac{3}{1 - 0.3 - 0.04} = 4.545 \Rightarrow \hat{y}_{100+h|100} \approx 4.545 \; for \; h = 100$$

Also, the variance of error is converging to unconditional variance:

$$Var(e_{t+100}) = \frac{\sigma^2}{1 - \phi_1^2 - \phi_2^2} = \frac{0.25}{1 - 0.09 - 0.0016} = 0.2752$$

$$\Rightarrow SE(e_{t+100}) = \sqrt{0.2752} = 0.525$$

4. Consider the following stationary and invertible $ARMA\,(1,1)$ model:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1},$$

for $t = 1, 2, \ldots, T$, where $\varepsilon_t \sim WN\,(0, \sigma^2)$. Suppose that the forecast origin is at time $t = T$.

(a) Find the 1-step-ahead forecast, given y_T and ε_T. (*Comment*: Recall that ε_T can be estimated using realizations $y_T, y_{T-1}, \ldots$, and hence we can consider it known at time T.)

(b) Show that starting from $h = 2$, the h-step-ahead forecast decays exponentially at the rate ϕ_1 toward the unconditional mean of $\{y_t\}$, as h increases.

(c) Show that the variance of the prediction error grows as we forecast further into the future.

(a)

$y_{T+1} = \phi_0 + \phi_1 y_T + \epsilon_{T+1} + \theta_1 \epsilon_T$

Using the fact that $\mathbb{E}[\epsilon_{T+1}|\Omega_T] = 0$

$\Rightarrow \hat{y}_{T+1} = \mathbb{E}[y_{T+1}|\Omega_T] = \phi_0 + \phi_1 y_T + \theta_1 \epsilon_T$

$e_{T+1} = y_{T+1} - \hat{y}_{T+1} = \epsilon_{T+1} \Rightarrow Var(e_{T+1}) = Var(\epsilon_{T+1}) = \sigma^2$

(b)

$y_{T+2} = \phi_0 + \phi_1 y_{T+1} + \epsilon_{T+2} + \theta_1 \epsilon_{T+1}$

Using the fact that $\mathbb{E}[\epsilon_{T+1}|\Omega_T] = 0$ and $\mathbb{E}[\epsilon_{T+2}|\Omega_T] = 0$

$\Rightarrow \hat{y}_{T+2} = \mathbb{E}[y_{T+2}|\Omega_T] = \phi_0 + \phi_1 \hat{y}_{T+1} = \phi_0 + \phi_0\phi_1 + \phi_1^2 y_T + \theta_1 \phi_1 \epsilon_T$

Continuing recursively, we have:

$$\hat{y}_{T+h} = \mathbb{E}[y_{T+h}|\Omega_T] = \phi_0 \sum_{i=0}^{h-1} \phi_1^i + \phi_1^h y_T + \phi_1^{h-1} \theta_1 \epsilon_T$$

$$= \frac{\phi_0(1 - \phi_1^h)}{1 - \phi_1} + \phi_1^h y_T + \phi_1^{h-1} \theta_1 \epsilon_T \; for \; h \geq 2$$

$\Rightarrow \lim_{h \to \infty} \hat{y}_{T+h} = \frac{\phi_0}{1-\phi_1}$ which is an unconditional mean of ARMA(1,1).

Thus, the forecast decays exponentially at rate ϕ_1 towards the unconditional mean.

(c)

$$e_{T+2} = y_{T+2} - \hat{y}_{T+2} = \phi_0 + \phi_1 y_{T+1} + \epsilon_{T+2} + \theta_1 \epsilon_{T+1} - (\phi_0 + \phi_1 \hat{y}_{T+1})$$
$$= \epsilon_{T+2} + (\phi_1 + \theta_1)\epsilon_{T+1}$$

Continuing recursively, we have:

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h} = \sum_{i=0}^{h-1} \phi_1^i (\epsilon_{T+h-j} + \theta_1 \epsilon_{t+h-j-1})$$

$$Var(e_{T+h}) = \sigma^2 \sum_{i=0}^{h-1} \phi_1^{2i}(1 + \theta_1^2) \Rightarrow \lim_{h\to\infty} Var(e_{T+h}) = \frac{\sigma^2(1+\theta^2)}{1-\phi_1^2}$$

Since the denominator is constant, and the numerator remains fixed, the variance increases with h but stabilizes at a finite bound. Thus, as we forecast further ahead, the uncertainty grows, but it does not diverge to infinity—it asymptotically approaches a steady-state level.